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ATOMIC CLUSTERS

A Study of Melting of Various Types of Pt–Pd Nanoparticles

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Abstract—The melting processes of various Pt–Pd nanoparticles (binary alloy, core–shell, $D \le 4.0$ nm) with different percent platinum atom content are investigated via the molecular dynamics using the embedded atom method potential in order to establish the thermal stability of simulated particle structure. In accordance with the data obtained, the most thermally stable are Pt–Pd nanoalloys with a diameter above 2.0 nm and core–shell Pd@Pt particles. As is shown, heating of binary Pt–Pd cluster alloys with the particle diameters less than 2.0 nm may cause the transition to pentagonal symmetry structures and core–shell-like complex formations.

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INTRODUCTION

Platinum is one of the most effective catalysts applied in various areas of engineering and chemical industry on account of its high catalytic activity [1]. However, its significant cost motivates searching for ways to increase the economic efficiency of Pt-catalysts. One such way is the elaboration of platinum-based multicomponent nanoparticles. For example, Pt–Au bimetallic nanoparticles exhibit better electrocatalytic activity and longer service life upon oxygen reduction, than single-metal platinum nanoparticles [2, 3]. It was also established that unlike Pt nanoclusters, bimetal Pt–Ru and Pt–Rh nanoparticles are stronger oxidants of CO [4, 5], etc.

Among the variety of platinum-based catalysts, one can distinguish the nanoalloy with palladium, which possesses lower cost and good catalytic activity and selectivity. Moreover, this nanoalloy is very promising as a catalyst in oxidation reactions and can be used in a low-temperature direct formic acid fuel cell [6].

It is evident that the catalytic activity of the nanoparticle, defined first of all of its surface properties, strongly depends on the structural properties of the latter, and the nanoparticle structure stability is thus critically important for catalysis. For bimetal Pt–Pd nanoparticles, there are three variations of the internal structural organization (Fig. 1): Pt-core/Pd-shell (Pt@Pd), Pd-core/Pt-shell (Pd@Pt) and a homogeneous particle of a bimetal alloy (Pt–Pd), at which all three kinds of nanoparticles are successfully synthesized (Fig. 2) [7–9].

On the other hand, it is well known that the chemical activity of Pt–Pd nanoparticles is on account of the size and structure, as well as on the platinum-to-palladium ratio therein [10]. In production of highly active nanocatalysts, it is therefore essential to take into consideration all peculiarities of the internal structure of such nanoparticles: diameter, core and shell dimensions, Pt and Pd atomic ratio, and surface structure. Since many industrial catalytic reactions occur at temperatures above 300–400°C, another aspect to be clarified is a careful study of melting–crystallization processes of nanoparticles that is of great importance in the determination of the degree of thermal stability of their structure.

To date, numerous experimental and theoretical works were mainly dedicated to various techniques of synthesis of bimetal Pt–Pd nanoparticles with different structures and their catalytic activity. Nevertheless, in our opinion, thermal evolution of structure and stability of these nanoparticles is still poorly understood. In this connection, the present work is aimed at elucidating the effect of thermal exposure on various palladium–platinum nanoparticles (Pt@Pd, Pd@Pt, Pt–Pd alloy, $D \le 4.0$ nm) via molecular dynamics.

SIMULATION TECHNIQUE

The molecular dynamics modeling was performed in the LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator) software, allowing us to successfully calculate the dynamic characteristics of a system simulated. This method has been widely used in recent years for studies of structural phase transformations in metals and alloys. A correct application of this type of modeling necessitates the accurate estab-

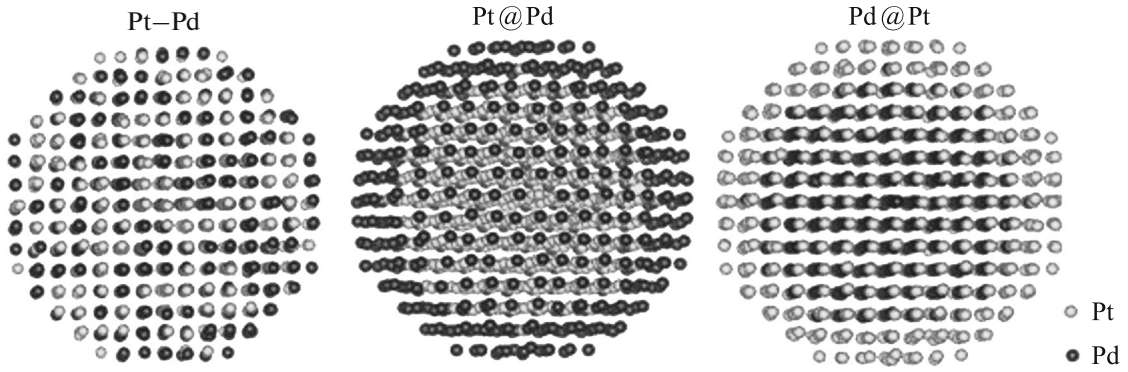


Fig. 1. The Pt–Pd nanoparticles with various structures (diameter of 3.0 nm): bimetal alloy (Pt–Pd), Pt-core/Pd-shell (Pt@Pd), and Pd-core/Pt-shell (Pd@Pt).

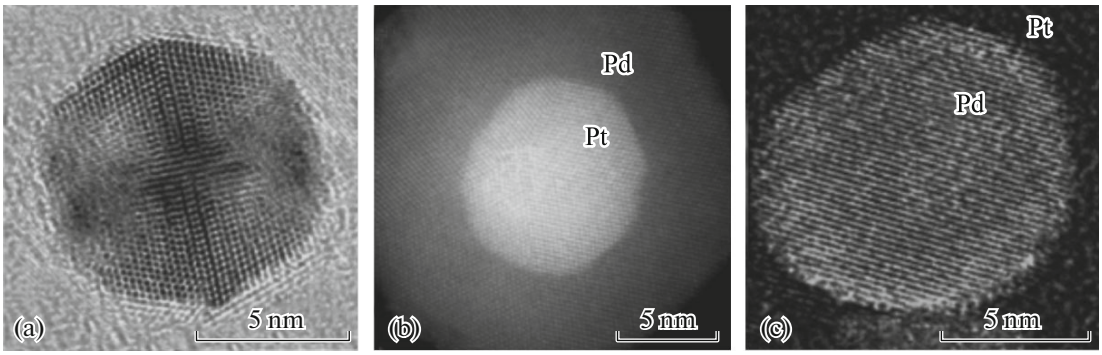


Fig. 2. TEM images of platinum and palladium nanoparticles with various structures. (a) Bimetal alloy Pt–Pd, (b) Pt-core/Pd-shell, (c) Pd-core/Pt-shell.

lishment of the interaction energy between the atoms and the system characteristics, required by the experimental conditions.

A careful analysis of references reveals that the interatomic interactions upon the MD simulation of thermodynamic properties of metal particles can be described by a group of multiparticle potentials, such as the Sutton–Chen, Cleri–Rosato, and the embedded atom method mechanisms (EAM) [11]. In this work, the interaction energy between platinum and palladium atoms was determined by means of the EAM potential proposed in [12], which was quite successfully applied for simulation of various nanoobjects of Pt–Pd alloy [13, 14].

The total energy (E_{tot}) in the approach we used was evaluated by the formula

$$E_{tot} = \frac{1}{2}\sum_{ij} V_{ij}(r_{ij}) + \sum_{i} F_i(\overline{p_i}), \quad (1)$$

where $V_{ij}(r_{ij})$ is the energy of interaction between atoms i and j at a distance r_{ij}, and $F_i(\overline{p_i})$ is the immersion potential of the i-th atom, depending on the effective electron density $\overline{p_i}$ in the vicinity of the atomic center.

The effective electron density was established as follows:

$$\overline{p_i} = \sum_{j\neq i} p_j(r_{ij}), \quad (2)$$

where $p_j(r_{ij})$ is the contribution to the electron density of the neighbor atom with a number j.

Another important criterion for successful MD modeling of thermal effect is the way to define the temperature in the system simulated. In a simulated experiment, the heating and cooling processes of nanoparticles can be implemented using a certain algorithm—thermostats. The most adequate in the task on determination of thermal stability of nanoparticles is considered a Nosé–Hoover thermostat [15–17], based on the immersion of a small system in a large heat reservoir. Herewith, the number of particles N, the volume V of a space simulated, and the temperature T are assumed to be constant, whereas the total pulse of a system p is equal to zero, which gives a so-called canonical or NVT ensemble.

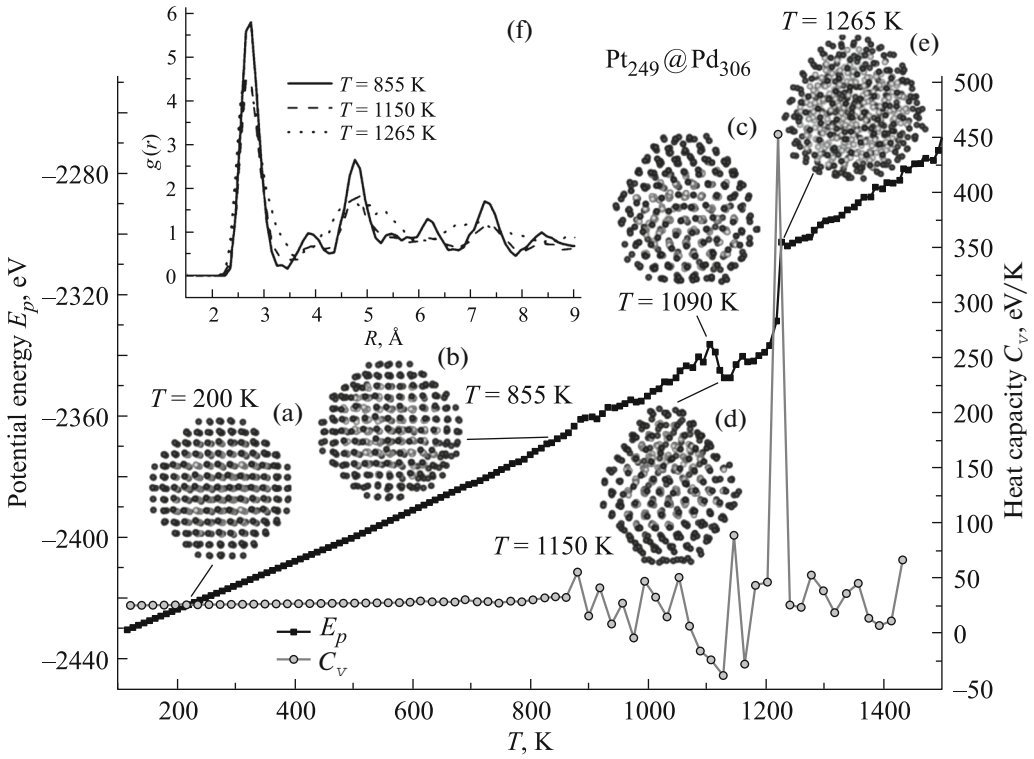


Fig. 3. The temperature dependence of the potential energy, heat capacitance and structure of a core–shell platinum–palladium nanoparticle ($Pt_{249}@Pd_{306}$).

The advantage of this technique is very good stabilization of a fixed temperature value with a maximum deviation of 1%. In our opinion, the Nosé–Hoover thermostat is most suitable for simulation of heating and cooling of isolated particles, where the essential aspect is the rigid fixation of kinetic energy of a system. The equations of particle motion in the system were solved by means of a Verlet method in its velocity form [18] with a constant time increment (h = 1 fs).

RESULTS AND DISCUSSION

To study the kinetics of possible thermoactivated structural modifications, the Pt–Pd particles with different structures were subjected to heating from T = 100 K to T = 2000 K for t = 20 ns. In core–shell Pt@Pd particles the Pt atom content varied from 10 to 50%. In particles with cores of Pd atoms and shells of Pt (Pd@Pt) atoms, the Pt atom concentration was from 50 to 90%. In bimetal Pt–Pd nanoparticles, an amount of Pt atoms achieved 10–90%.

To elucidate structural transitions and melting processes, the potential energy of the particle was fixed, and the radial distribution function and heat capacitance were calculated at each step of simulation. The typical temperature dependences of these parameters are shown in Fig. 3 for a $Pt_{249}@Pd_{306}$ particle. A visual inspection of these plots highlights small fluctuations in potential energy and heat capacitance at a temperature of ~850 K. With a further increase in temperature to T = 1090 K there is an essential diffusion rearrangement of atomic positions in the particle, leading to the formation of facets on its surface, and at T = 1150 K the particle finally takes a rhombic shape that is accompanied with a decrease in the potential energy, on account of the reduction in its surface component. We mention that the formation of such corelike rhombic and triangle nanoparticles is also observed experimentally (Fig. 4).

Despite the change in the external shape, the order of atomic arrangement in $Pt_{249}@Pd_{306}$ particle remains almost unchanged with increased temperature that is evident from the radial distribution function (the insert in Fig. 3).

So, the distribution peaks remain at the same distances when augmenting the temperature from 855 to 1150 K, which confirms the unchanged crystal structure, and a decrease in their maximum values is caused by increased heat fluctuations of stable equilibrium positions of atoms composing a nanoparticle. At a further increase in the temperature, one observes the particle melting along with the characteristic jumps in the potential energy and heat capacitance (Fig. 3).

Another situation was observed upon the heating of platinum–palladium nanoparticles with a reverse structure Pd@Pt. Figure 5 displays the typical temperature dependence of the potential energy, heat

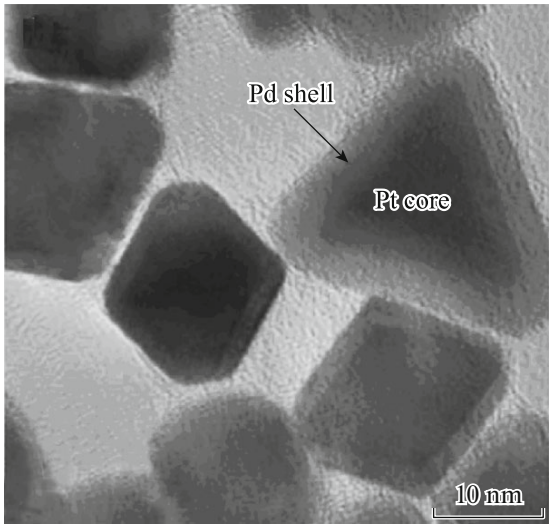


Fig. 4. TEM images of Pt–Pd nanoparticles with various shapes synthesized via chemical methods (using the data [22]).

capacitance, and structure of $Pd_{55}@Pt_{500}$ nanoparticles. Similarly, with a cluster $Pt_{249}@Pd_{306}$ with the identical diameter, one observed the variations in the potential energy and heat capacitance at a temperature of 850 K that was referred to the internal rearrangements of the cluster. On the other hand, no drastic changes in structure and particle shape were detected until the melting point. Nevertheless, the particle melting is followed by intensive diffusion of Pd atoms to the surface that decreases the potential energy a little bit.

A similar situation occurs also upon thermal exposure of nanoparticles with a bimetal Pt–Pd alloy (Fig. 6). According to the modeling data, the heating exerts a small influence on the internal structure of nanoalloy for particles with D = 2.0–4.0 nm at any palladium content. Thus, the Pt–Pd nanoparticles preserved the initially defined fcc structure up to the phase transition point (crystal–liquid), and a loss of long-range order was only after its completion.

The modeling experiment revealed no regularities in arrangement of first- or second-type particles in bimetal Pt–Pd clusters. The impurity metal particles were randomly distributed at any temperature in the system simulated, being in both the upper layer of a nanocluster and at its center. Thus, the heating of Pt–Pd nanoclusters with a diameter of D = 2.0–4.0 nm up to the melting point caused neither decomposition of chemical elements, nor formation of more complex core–shell or janus structures. This conclusion is confirmed by the experimental study [19], where the analysis of the internal structure of chemically synthesized Pt–Pd nanoparticles with a diameter of 11.2–0.8 nm

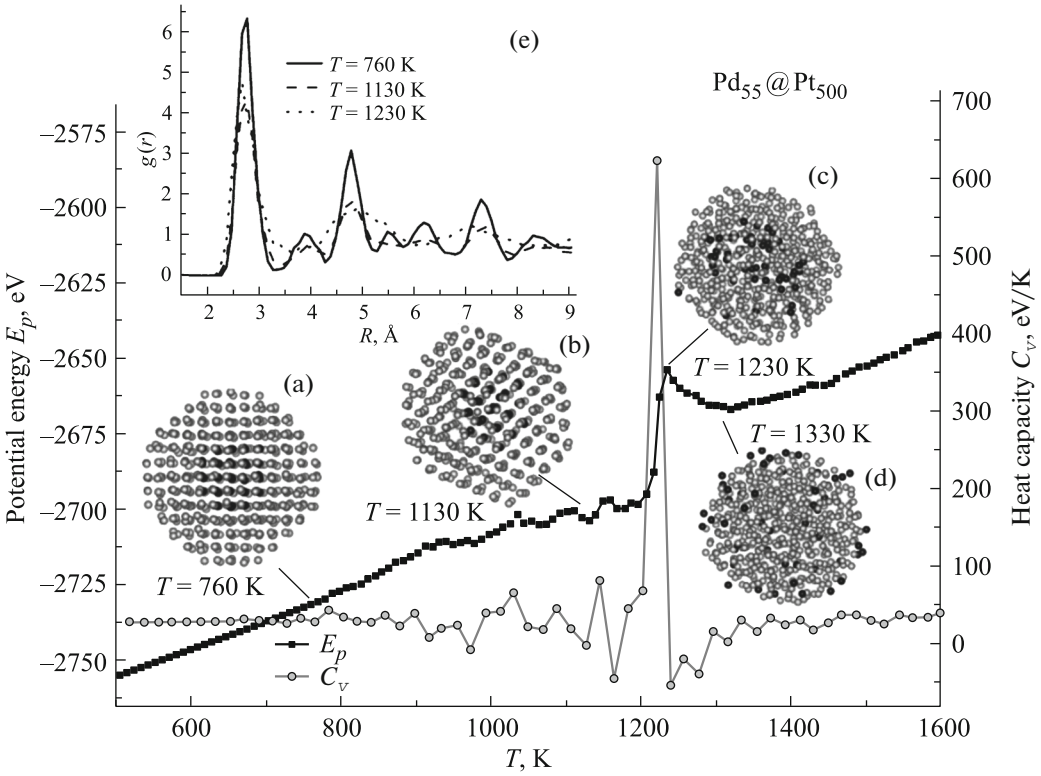


Fig. 5. The temperature dependence of the potential energy, heat capacitance and structure of a core–shell platinum–palladium nanoparticle ($Pd_{55}@Pt_{500}$).

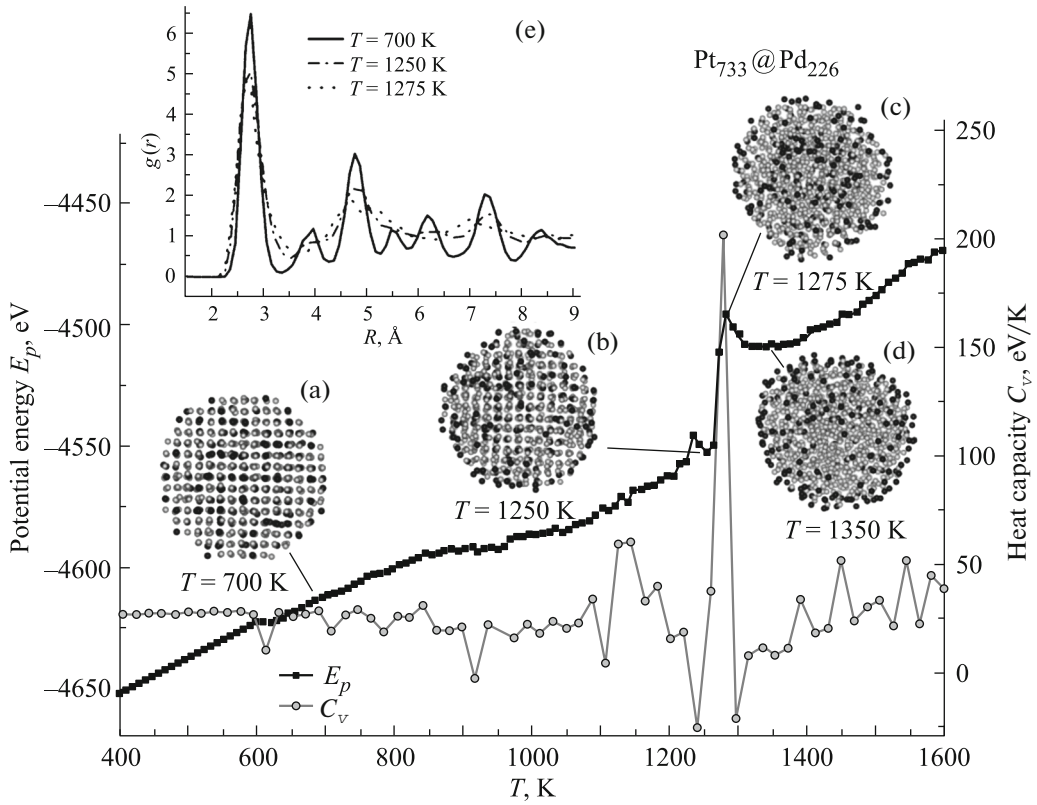


Fig. 6. The temperature dependence of the potential energy, heat capacitance and structure of a bimetal platinum–palladium nanoparticle ($Pt_{733}Pd_{226}$).

highlighted that the platinum and palladium atoms in most cases for the Pt–Pd nanoalloy with randomly distributed particles. However, after melting even these particles undergo intensive diffusion of Pd atoms to the surface by analogy with core–shell Pd@Pt systems (Fig. 6).

Although synthesis of bimetal Pt–Pd particles with a diameter less than 2.0 nm is quite a complex chemical task, it is, however, possible, if using the ALD (atomic layer deposition) technology [20]. In this work, the Pt–Pd particles with diameters of 1.3–0.3 nm and 1.7–0.5 nm were obtained experimentally, but the electron microscopy methods could not provide the reliable structure of the above particles because of the smallness of the particles. In this connection, the clusters of Pt–Pd alloys with a number of particles of 135 atoms and the above diameters were hereinafter studied via simulation.

In a careful investigation of changes in the internal structure of such a fine nanoalloy, one could distinguish some basic peculiarities of the behavior of a nanoparticle, in particular, several probable scenarios of thermal evolution of bimetal Pt—Pd alloy (N = 135 atoms).

The first of them corresponds to the behavior characteristic of the above studied nanoalloy particles with D = 2.0–4.0 nm. Upon heat supply, the potential energy of a cluster increases gradually to the melting point with retained initial fcc structure of a nanoparticles that is established for evolution of nanoclusters which contained up to 5% of platinum atoms.

The second scenario of thermal evolution manifests itself in a fcc → pentagonal symmetry structural transition during the heating that was highlighted for a Pt–Pd nanoalloy with 10 to 35% platinum atoms that coincides with the data reported in [21]. So, the calculation of the ground state energy for a Pt–Pd cluster (N = 98 atoms) in [21] revealed that the energy minimum is approximately at 33–50% of platinum content in a bimetal particle, and in many cases, namely, the decaedric structure was found to be the most stable.

The third scenario of the nanoalloy behavior highlighted great enough fluctuations in the potential energy caused by heating of particles, but no significant changes in the alloy geometry were observed. It occurred at 40–80% Pt concentration in a bimetal alloy.

The ultimate scenario of thermal evolution of a bimetal Pt–Pd lloy (N = 135 atoms at the palladium content to 20%) possessed a curious property. Before the complete structural melting and displacement of palladium atoms in the surface layer, the potential energy abruptly increased that induced structural rearrangement towards the pentagonal symmetry.

With further heating, the formed structures collapsed, passing in the amorphous state, but already with palladium in the surface layer.

CONCLUSIONS

Platinum-based nanoalloys attract high attention of researchers due to prospects of their active use in chemical industry, electrocatalysis, microelectronics, medicine, and other fields. However, to achieve the highest catalytic activity, these particles have to possess a complex structure, such as core–shell, crown-like, or multiple twins. Since the activity and selectivity of catalytic structures with the use of nanoparticles is closely related to the internal structure, structural stability of nanoparticles is often the determining factor for their industrial application. Nevertheless, gathering this information directly from the experimental data is difficult, because of smallness of the particles studied, and thus these aspects are still poorly understood.

In the present work, the kinetics of possible thermoactivated structural rearrangement of Pt–Pd nanoparticles with different structure and Pt-to-Pd atomic ratios was characterized via molecular dynamics modeling using the embedded atom method potential. As was established, the structural type of nanoparticles exerts a strong influence on their structural stability upon thermal exposure. For Pt-core and Pd-shell nanoparticles, the heat energy supply may cause severe structural changes much below the melting point with a formation of the rhombic shape which was observed experimentally during the synthesis of bicomponent Pt–Pd nanoparticles.

In the case of Pd-core, Pt-shell and bimetal Pt–Pd alloy nanoparticles the heating led to no significant structural changes up to the temperature of transition into liquid state. On the other hand, the melting in such particles was followed with intensive diffusion of Pd atoms to the surface.

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